Rheumatoid arthritis

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Rheumatoid arthritis is a chronic, systemic, autoimmune inflammatory disease that mainly affects the joints and periarticular soft tissues. In this Seminar, we provide an overview of the main aspects of rheumatoid arthritis. Epidemiology and advances in the understanding of rheumatoid arthritis pathogenesis will be reviewed. We will discuss the clinical manifestations of rheumatoid arthritis, classification criteria, and the value of imaging in the diagnosis of the disease. The advent of new medications and the accumulated scientific evidence demand continuous updating regarding the diagnosis and management, including therapy, of rheumatoid arthritis. An increasing number of patients are now able to reach disease remission. This major improvement in the outcome of patients with rheumatoid arthritis has been determined by a combination of different factors (eg, early diagnosis, window of opportunity, treat-to-target strategy, advent of targeted disease-modifying antirheumatic drugs, and combination therapy). We will discuss the updated recommendations of the two most influential societies for rheumatology worldwide (ie, the American College of Rheumatology and European Alliance of Associations for Rheumatology) for the management of rheumatoid arthritis. Furthermore, controversies (ie, the role of glucocorticoids in the management of rheumatoid arthritis and safety profile of Janus kinase inhibitors) and outstanding research questions, including precision medicine approach, prevention, and cure of rheumatoid arthritis will be highlighted.

Published **Online**
October 27, 2023
https://doi.org/10.1016/S0140-6736(23)01525-8

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Introduction

Rheumatoid arthritis is a systemic autoimmune inflammatory disease that primarily affects the joints. Rheumatoid arthritis is associated with progressive disability, premature death, and high socioeconomic costs.[1,2] The prevalence of rheumatoid arthritis varies substantially around the globe, ranging from 0·25% to 1%.[3] Rheumatoid arthritis can affect individuals of any age, with an increased incidence in people older than 40 years. Women are affected two to three times more frequently than men.[4]

Disease susceptibility and pathogenesis

The pathogenesis of rheumatoid arthritis is complex and not fully understood.[5,6] Genetic predisposition plays a key role, with first-degree relatives of patients with rheumatoid arthritis carrying a two to five times higher risk of developing the disease compared with the general population.[7] Multiple alleles have been associated with an increased risk of rheumatoid arthritis. The strongest evidence supports the role of *HLA-DRB1* gene, and in particular a key sequence of five amino acid sequence motif in residues 70–74 of the HLA-DRβ chain, the so-called shared epitope.[8,9] Several non-MHC genes have been linked with rheumatoid arthritis (eg, *PTPN22*, *CTLA4*, and *PADI4*), but their individual contribution is modest compared with the HLA region.[10,11] Large-scale genome-wide association studies have identified numerous genetic mechanisms and loci associated with rheumatoid arthritis, such as *TNIP2*, *WISP1* and *TNFRSF11A*.[12]

Epigenetic factors, such as DNA methylation and histone acetylation, also contribute to rheumatoid arthritis pathogenesis.[13] In genetically predisposed individuals, environmental risk factors, such as smoking (ie, the strongest environmental risk factor for rheumatoid arthritis), dust exposure, viruses, obesity, low socioeconomic status, and changes in the lung, gut, and oral microbiome, have been implicated in the so-called break of tolerance (ie, the production of autoantibodies and generation of proinflammatory cytokines), which appear to bring about disease initiation and perpetuation.[14–16] Conversely, a moderate alcohol intake seems to be protective.[17] A study has shown an increasing incidence of rheumatoid factor (RF)-negative rheumatoid arthritis compared with previous decades.[18] However, the precise interplay between genetic and environmental risk factors and how this interaction leads to the eventual development of the disease have not been fully elucidated.

The production of autoantibodies and the presence of autoreactive T cells in blood and synovial structures are distinguishing features of rheumatoid arthritis. Anti-citrullinated protein antibodies (ACPAs) and, to a lesser extent, RF have shown the strongest association with rheumatoid arthritis.[19,20] Other autoantibodies have been linked to rheumatoid arthritis, including peptidyl arginine deiminase-4 antibodies, carbamylated protein antibodies, and collagen type 2 antibodies.[21,22]

Rheumatoid arthritis-related autoimmunity precedes the onset of clinical disease by up to several years.[23] Observations suggest that, in some patients, the initial site of autoantibody generation is distant from the joint, specifically the mucosae of the periodontium, gut, or lung.[24] This initial phase of localised mucosal autoimmunity (ie, IgA-related) would be then followed by transition to systemic autoimmunity (ie, IgG-related) and expansion of the numbers and specificities of ACPAs, leading to a complex proinflammatory immune response and eventually disease initiation and perpetuation.[25]

Among the different rheumatoid arthritis-related autoantibodies, ACPAs and, in particular, the anti-cyclic citrullinated peptide (CCP) antibodies have shown the highest specificity for rheumatoid arthritis.[26] ACPAs have an important diagnostic role and are of potential

prognostic value, as their presence has been associated with radiographic progression, extra-articular manifestations, and response to treatment.[27]

From a histological point of view, the rheumatoid pannus represents the hallmark sign of rheumatoid arthritis. This pannus has hyperplasia of the synovial tissue, neovascularisation, and a heterogeneous inflammatory cell infiltrate of activated $CD4^+$ $CD8^+$ T cells, B cells, immunoglobulin-producing plasma cells, macrophages, fibroblasts, neutrophils, and dendritic cells.[28,29] If left untreated, this highly aggressive pannus invades adjacent structures, leading to progressive and irreversible damage of bone, cartilage, tendons, and ligaments. Activated osteoclasts and chondrocytes play a direct role in the development of joint structural damage.[30] Several mediators have been identified in the interaction between the cells involved in this inflammatory network, such as tumour necrosis factor (TNF), IL-6, IL-1 and transforming growth factor beta (TGF-β).[31]

Clinical manifestations

In most patients, the clinical presentation of rheumatoid arthritis is an insidious and gradual onset of joint pain and swelling, which could start with one or a few joints but usually develops into a symmetrical polyarthritis.[32] The wrists and small joints of the hands and feet are the most affected joints. Any synovial joint, including large joints (eg, shoulders, elbows, knees and ankles) can also be involved. Early morning stiffness lasting 30 min or longer, fatigue, and weakness are common features. In patients with severe systemic inflammation, such as those with markedly elevated C-reactive protein and erythrocyte sedimentation rate, anorexia, weight loss, and low-grade fever might occur.

In the early phases of rheumatoid arthritis, joint function is mainly impaired by stiffness (or gelling) and pain, which are secondary to active inflammatory process in both the joints and periarticular soft tissues, such as tendons, including the recently described peritendinitis of the interossei.[33] If not properly treated with disease-modifying antirheumatic drugs (DMARDs), this inflammatory process progresses to the development of irreversible joint damage, with consequent loss of function (figure 1).

One or more extra-articular manifestations, such as subcutaneous nodules, occur in about 40% of patients with rheumatoid arthritis, especially those with long-standing disease.[34,35] Subcutaneous nodules, also known as rheumatoid nodules, are mainly observed in patients with positive anti-CCP antibodies or RF and are exacerbated by smoking. Other common extra-articular manifestations are haematological abnormalities (eg, anaemia, thrombocytosis, and lymphadenopathy) and interstitial lung disease, and less common and rare manifestations include vasculitis, neuropathy, serositis, glomerulonephritis, inflammatory eye involvement, Felty syndrome, myopathy, and amyloidosis. The prevalence of many of these extra-articular manifestations has decreased over time, presumed to be due to better and earlier treatment of inflammation.

Diagnosis and classification of rheumatoid arthritis

To date, no diagnostic criteria exist for rheumatoid arthritis. In routine practice, many rheumatologists rely on the fulfilment of the American College of Rheumatology (ACR) and European Alliance of Associations for Rheumatology (EULAR) classification criteria for rheumatoid arthritis, which replaced the previous 1987 ACR criteria in 2010.[36,37]

The entry criterion is at least one clinically swollen joint. Other types of arthritis, such as crystal arthritis, reactive arthritis, and connective tissue diseases must be excluded to apply these criteria. The main domains of these criteria are joint distribution (ie, large joints *vs* small joints), serological status (ie, positive ACPA or RF), acute phase reactants, and duration of symptoms. In 2014, a systematic literature review showed a pooled sensitivity of 0·82 (95% CI 0·79–0·84) and specificity of 0·61 (0·59–0·64) for the 2010 ACR and EULAR classification criteria.[38]

Imaging is increasingly used in clinical practice to identify inflammation and early structural damage. Ultrasonography, MRI, and conventional radiography are useful when the diagnosis of rheumatoid arthritis is in question. Ultrasonography has been proven to be sensitive and accurate for the detection of synovial hypertrophy and pathological vascularisation (ie, power doppler signal) in the joints and tendons of patients with rheumatoid arthritis.[39]

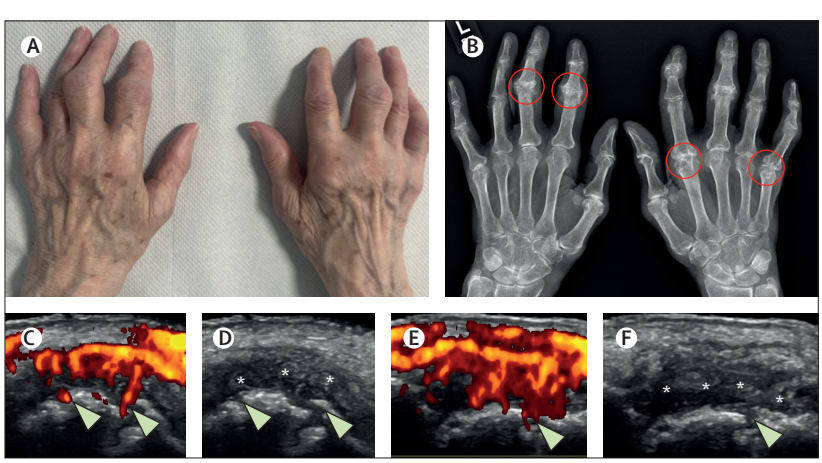


Figure 1: **Long-standing rheumatoid arthritis**
(A) Polyarticular synovitis and joint deformities are shown in the hands of a patient with long-standing rheumatoid arthritis (ie, >20 years disease duration). (B) X-rays show diffuse structural damage (ie, severe cartilage loss and bone erosions in multiple metacarpophalangeal and proximal interphalangeal joints bilaterally), with complete loss of normal joint architecture (ie, ulnar deviation, joint subluxation, and multiple bone erosions) in the metacarpophalangeal and proximal interphalangeal joints (red circles). A longitudinal ultrasonography scan of the medial aspect of the second right metacarpophalangeal joint (C, D) and second left metacarpophalangeal joint (E, F) shows diffuse synovial hypertrophy (asterisks) and large bone erosions (arrows) in the metacarpal head filled with power doppler signal (intrasynovial red spots), indicating active synovitis (ie, bone erosions).

MRI can detect synovitis, tenosynovitis, and osteitis, which are potential predictors of the development of bone erosions.[40] Conventional radiography is the first-line imaging test for the detection of joint damage according to EULAR.[41] Ultrasonography and MRI represent promising tools for the assessment of structural joint damage due to their higher sensitivity compared with conventional radiography (figure 2).[42] The value of imaging in the follow-up of patients with rheumatoid arthritis has not been clearly established, with some studies not showing the additional value of ultrasonography over clinical measures in the treatment decision making of patients with rheumatoid arthritis.[43,44]

Treatment

Timely intervention has a disproportionate long-term impact and the prompt initiation of DMARD therapy within a few weeks or months of disease onset has been proven crucial to improving prognosis and long-term outcomes in rheumatoid arthritis.[45] The precise timing for this opportunity has not yet been defined, but is assumed to be 3 months; nevertheless, there is unequivocal evidence showing that initiation of DMARD therapy early in the course of the disease decreases structural damage and progression and improves physical functioning compared with delayed treatment.[46]

Three categories of DMARDs constitute the therapeutic armamentarium of rheumatoid arthritis: (1) conventional synthetic DMARDs, which are small-molecular-weight synthetic drugs (ie, small chemical drugs) with unclear anti-inflammatory mechanisms; (2) biological DMARDs, which are mostly monoclonal antibodies or, less often, soluble receptor constructs, which specifically target an individual molecule; and (3) targeted synthetic DMARDs, which target specific enzymes within cells (table). Among the conventional synthetic DMARDs, methotrexate and, to a lesser extent, sulfasalazine, leflunomide, and hydroxychloroquine are the most commonly used drugs in rheumatoid arthritis. Methotrexate is traditionally regarded as the anchor drug (ie, the first-line therapy, to which other drugs are added) in rheumatoid arthritis.[47]

The advent of biological DMARDs has substantially changed the natural history of rheumatoid arthritis in the majority of patients.[48] Multiple biological DMARDs with different mechanisms of action have been developed, ranging from direct inhibition of TNF-α and IL-6 receptor antagonism to blockade of T-cell co-stimulation and B-cell depletion. The safety profile of the biological DMARDs has now been established (table), with most of these drugs sharing an increased risk of infections as the most common side-effect. In addition, targeted synthetic DMARDs inhibiting the Janus kinase (JAK) pathways are also licensed for the treatment of rheumatoid arthritis. Several randomised controlled trials showed a greater efficacy of combinations of biological DMARDs or targeted synthetic DMARDs plus conventional synthetic DMARDs versus a biological DMARD, targeted synthetic DMARD, or conventional synthetic DMARD alone.[49,50] One action of methotrexate is reducing the development of antidrug antibodies, particularly those against chimeric antibodies.[51] Specific less immunogenic monoclonal antibodies, such as some TNF inhibitors or IL-6 inhibitors, are licensed as monotherapy.[52] For some of the original biological DMARDs, for which patents have expired, less expensive biosimilars have been developed and approved as effective, safe, and equivalent to the bio-originators.

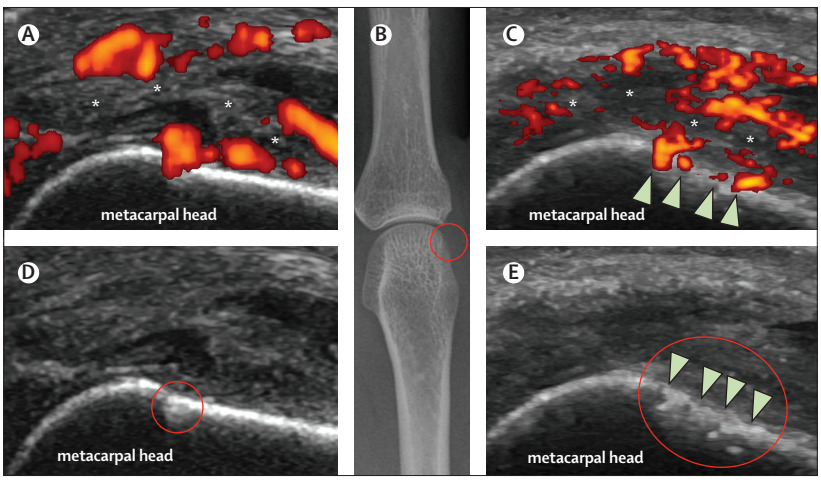


Figure 2: **Conventional radiography and ultrasonography findings in early rheumatoid arthritis**
Longitudinal ultrasonography (A, C–E) and x-rays (B) of the right third metacarpophalangeal joint in a patient with early rheumatoid arthritis (ie, disease duration of 16 weeks). The ultrasonography images, which were done with a 15 MHz (A, D) and 18 MHz (C, E) probe, show synovial hypertrophy (A, C; asterisks) and diffuse power doppler signal (D, E; red circles), indicating active inflammation. High-frequency probes (ie, 18 MHz probe) provide a detailed morphological evaluation of the superficial structures. In this patient, loss of sharpness of the bony cortex, suggestive for a pre-erosive change, can be seen (E; arrows). The presence of highly vascularised synovial hypertrophy (C, E; asterisks and red circles) is visible in close contact with the bone surface (C; arrows). No abnormality was reported in the x-rays of this patient (D).

ACR and EULAR recommendations for rheumatoid arthritis management

The two most influential organisations for rheumatology worldwide updated versions of recommendations and guidelines for the management of rheumatoid arthritis, the ACR in 2021[53] and EULAR in 2022[54] (figure 3). Both the ACR and EULAR highlight the importance of the window of opportunity, which states that DMARDs should be started as soon as the diagnosis of rheumatoid arthritis is made, and the treat-to-target strategy, which consists of tight disease activity control with regular follow-up of patients (every 1–3 months according to EULAR) and prompt treatment escalation if remission is not attained. In some patients, in whom disease remission is not possible, low disease activity would be an acceptable target in place of remission.

First-line therapy should be a conventional synthetic DMARD, most commonly methotrexate, because of insufficient evidence supporting the higher efficacy of biological DMARDs in all patients and the higher costs of these medications compared with methotrexate.

Although EULAR recommends methotrexate in all newly diagnosed patients with rheumatoid arthritis unless contraindicated, the ACR considers the possibility of different conventional synthetic DMARDs according to disease activity. In patients who are DMARD-naive and have moderate-to-high disease activity, methotrexate is strongly recommended over other conventional synthetic DMARDs, such as hydroxychloroquine or sulfasalazine. By contrast, in those with low disease activity, hydroxychloroquine is conditionally recommended over other conventional synthetic DMARDs because of tolerability and safety profiles. The ability to predict

	Target	Route of administration (dose)	Adverse events	Management of rheumatoid arthritis
Glucocorticoids	Genomic and non-genomic pathways	Oral, intramuscular, intravenous, intra-articular (dose depends on route of administration and clinical indication)	Depending on dose and duration of therapy; diabetes, hypertension, infections, gastrointestinal diseases, skin atrophy, psychiatric, osteoporosis, and ophthalmological diseases (eg, cataract and glaucoma)	Bridging therapy when DMARD therapy is initiated or switched and treating flares; in some patients is sometimes used as a long-term maintenance therapy (similar to a DMARD)
Conventional synthetic DMARDs	..	..	..	First-line therapy in patients who are naive to DMARDs
Methotrexate	Unknown	Oral, subcutaneous, intramuscular (10–25 mg/week)	Gastrointestinal, increased liver enzymes, bone marrow suppression, stomatitis, hair loss, teratogenicity, and interstitial pneumonitis	First choice among conventional synthetic DMARDs
Sulfasalazine	Unknown	Oral (2–3 g/day)	Gastrointestinal, skin rash, temporary oligospermia, and drug-induced systemic lupus erythematosus	Combination therapy with methotrexate (or monotherapy if methotrexate is contraindicated)
Leflunomide	Dihydroorotate dehydrogenase	Oral (20 mg/day)	Gastrointestinal, increased liver enzymes, leukopenia, hypertension, and teratogenicity	Monotherapy if methotrexate is contraindicated (combination therapy with methotrexate uncommon)
Hydroxychloroquine	Unknown	Oral (200–400 mg/day)	Gastrointestinal, skin rash, and retinopathy	Combination therapy with methotrexate (or monotherapy if methotrexate is contraindicated in patients with low disease activity)
Targeted synthetic DMARDs (ie, JAK inhibitors)	..	..	Gastrointestinal, infections, colonic perforation, increased creatinine kinase concentrations, tuberculosis or herpes zoster reactivation, cytopenia, increased risk of venous thromboembolism, and lipid abnormalities	In patients who have had at least one conventional synthetic DMARD, after at least one TNF inhibitor (ACR), or as first-line therapy (EULAR) in selected populations*; might have some advantages in monotherapy compared with other biological DMARDs
Tofacitinib	JAK 1,2,3	Oral (10 mg/day)	Possible increased risk of major adverse cardiovascular event and neoplasm (especially in smokers and people older than 65 years)	..
Baricitinib	JAK 1,2	Oral (2–4 mg/day)	..	..
Upatacitinib	JAK 1,2†	Oral (15 mg/day)	..	..
Filgotinib	JAK 1	Oral (200 mg/day)	..	..
Biological DMARDs	..	..	..	First-line therapy in patients who have had at least one unsuccessful conventional synthetic DMARD
TNF inhibitors	..	..	Infections, tuberculosis reactivation, cytopenia, drug-induced systemic lupus erythematosus, non-melanoma skin cancer, demyelinating syndromes, congestive heart failure, and infusion or injection-related reactions	Commonly used as first-line therapy among biological DMARDs
Adalimumab	TNF	Subcutaneous (40 mg every 2 weeks)	..	..
Etanercept	TNF	Subcutaneous (50 mg/week)	..	..
Golimumab	TNF	Intravenous (2 mg/kg at week 0, 4, and every 8 weeks); subcutaneous (50 mg every 4 weeks)	..	..
Certolizumab pegol	TNF	Subcutaneous (400 mg at week 0, 2, and 4, then 200 mg every 2 weeks)	..	..
Infliximab	TNF	Intravenous (3 mg/kg at week 0, 2, and 6, and every 8 weeks); subcutaneous (120 mg every 2 weeks)	..	..

(Table continues on next page)

	Target	Route of administration (dose)	Adverse events	Management of rheumatoid arthritis
(Continued from previous page)				
Anti-B cell	..	..	..	In patients, who have had at least one conventional synthetic DMARD, usually after TNF inhibitors; ACR suggests use after inadequate response to TNF inhibitors or in patients with history of lymphoproliferative disorder
Rituximab	CD20	Intravenous (1–2 g every 6 months)	Infusion reactions, reduced response to vaccines, infections, hepatitis B reactivation, and progressive multifocal leukoencephalopathy	..
Anti T-cell co-stimulation	..	..	..	First-line therapy in patients who have had at least one unsuccessful biological or conventional synthetic DMARD
Abatacept	CD80–CD86 co-stimulation	Intravenous (500–1000 mg, depending on weight, at week 0, 2, and 4 and then every 4 weeks); subcutaneous (125 mg/week)	Infections, infusion or injection site reaction, reduced response to vaccines, tuberculosis reactivation, and leukopenia	..
Anti-IL-6	..	..	Infections, colonic perforation, lipid abnormalities, cytopenia, liver enzyme elevations, and infusion or injection-related reactions	First-line therapy in patients who have had at least one conventional synthetic DMARD; might have some advantages compared with biological DMARDs in monotherapy
Tocilizumab	IL-6 receptor	Intravenous (4–8 mg/kg every 4 weeks, max 800 mg); subcutaneous (162 mg/week)	..	..
Sarilumab	IL-6 receptor	Subcutaneous (150–200 mg every 2 weeks)	..	..

Anakinra is approved for the treatment of rheumatoid arthritis in the USA and Europe; however, clinical trials showed modest therapeutic effects in rheumatoid arthritis, and it is not mentioned in EULAR and ACR recommendations. ACR=American College of Rheumatology. DMARD=disease-modifying antirheumatic drug. EULAR=European Alliance of Associations for Rheumatology. JAK=Janus kinase. TNF=tumour necrosis factor. *Risk factors for cardiovascular events and malignancies to consider before prescribing a JAK inhibitor: age older than 65 years, previous or current smoking, diabetes, obesity, hypertension, current or previous malignancy (other than non-melanoma skin cancer), and risk factors for thromboembolic events (eg, history of myocardial infarction or heart failure, history of blood clots or inherited disorders of coagulation, combined contraceptives or hormonal replacement therapy, immobility, and undergoing major surgery). †Various assays showed different specificities for JAK 2 inhibition.

Table: **Main characteristics of glucocorticoids, conventional synthetic DMARDs, biological DMARDs, and targeted synthetic DMARDs, which are approved for the treatment of rheumatoid arthritis in the USA and Europe**

methotrexate response at baseline would improve management. There are some positive developments in this area that could alter management, including a proof-of-concept study (NCT03813771) nearing completion, which might provide clarification. In patients with an insufficient response to a conventional synthetic DMARD, risk stratification is clinically sound, although the evidence for treating patients according to prognostic markers is inadequate. EULAR recommends the use of other conventional synthetic DMARDs in patients without poor prognostic factors (eg, autoantibodies, high disease activity, bone erosion, and use of two ineffective conventional synthetic DMARDs). Conversely, in those with poor prognostic factors, a biological DMARD (or a targeted synthetic DMARD in selected populations) should be added. The ACR does not provide any specific indication regarding the sequence of these medications in patients with persistently active rheumatoid arthritis despite conventional synthetic DMARD therapy, but recommends the addition of a biological or targeted synthetic DMARD over triple therapy (ie, methotrexate, sulfasalazine, and hydroxychloroquine). Importantly, local regulations vary, with mandatory treatment attempts with two different conventional synthetic DMARDs required in some countries (eg, the UK) before starting a biological DMARD.

The role of glucocorticoids in rheumatoid arthritis management

Glucocorticoids have traditionally been represented as fundamental in the management of rheumatoid arthritis. Given their availability in different doses and formulations (ie oral, parenteral, or intra-articular) and the rapid onset of anti-inflammatory activity, glucocorticoids are commonly used as a bridging therapy when DMARD therapy is initiated, for treating flares, and, in some patients, as a long-term maintenance therapy.[55]

Multiple studies have shown the ability of glucocorticoids to improve disease activity and functional status in patients with rheumatoid arthritis, and potentially delay radiographic joint damage (ie, like DMARDs).[56,57] The NORD-STAR trial, which included 812 patients who had rheumatoid arthritis for more than 2 years and were treatment naive, showed non-inferiority of conventional synthetic DMARDs plus glucocorticoids compared with certolizumab pegol with methotrexate and tocilizumab with methotrexate, and abatacept with methotrexate was statistically superior.[58] In addition, NORD-STAR showed clinical similarity between conventional synthetic DMARDs with glucocorticoid therapy and any biological DMARD with methotrexate treatment, with high rates of stringent remission in Clinical Disease Activity Index at 24 weeks (>40%) for all

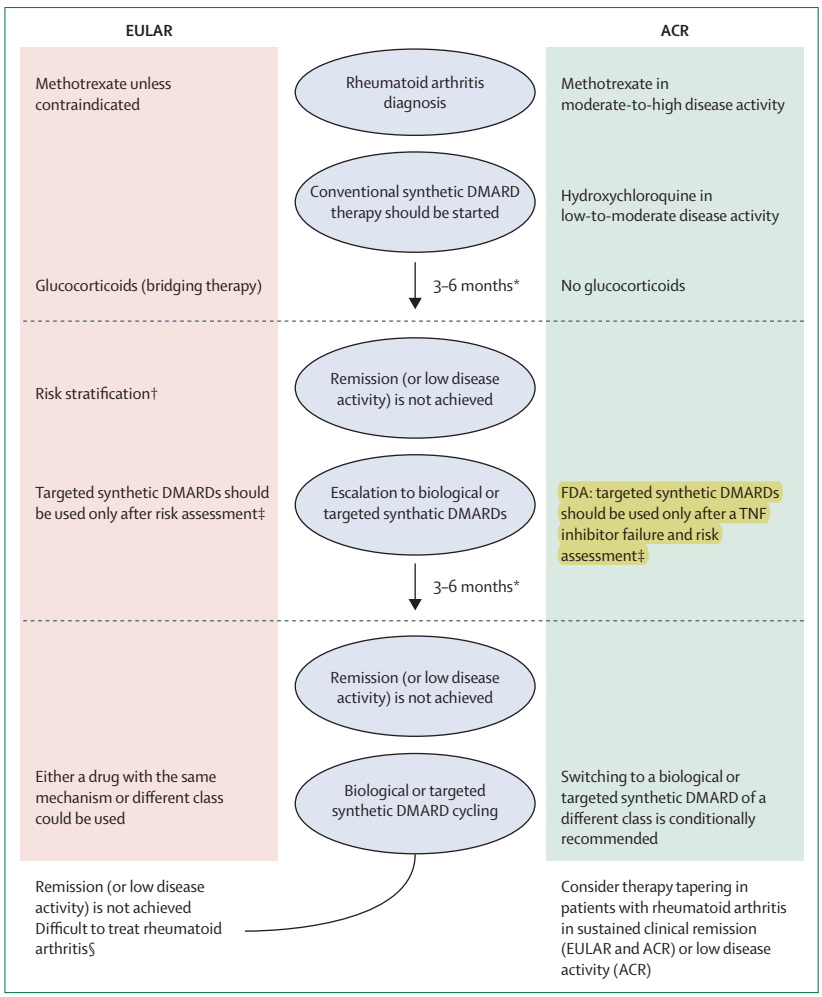


Figure 3: **The therapeutic management of rheumatoid arthritis according to ACR and EULAR guidelines and recommendations**

ACR=American College of Rheumatology. DMARD=disease-modifying antirheumatic drugs. EULAR=European Alliance of Associations for Rheumatology. FDA=US Food and Drug Administration. TNF=tumour necrosis factor. *Treatment target is remission (according to the ACR and EULAR definition) or low disease activity at 6 months. According to EULAR, therapy should be adapted if 50% improvement is not reached at 3 months. †In the presence of poor prognostic factors (eg, autoantibodies, high disease activity, bone erosion, and two ineffective conventional synthetic DMARDs), a biological or targeted synthetic DMARD should be added. Otherwise, another conventional synthetic DMARD should be considered. ‡Risk factors for cardiovascular events and malignancies to consider before prescribing a Janus kinase inhibitor are age older than 65 years, previous or current smoking, diabetes, obesity, hypertension, current or previous malignancy (other than non-melanoma skin cancer), and risk factors for thromboembolic events (eg, history of myocardial infarction or heart failure, history of blood clots or inherited disorders of coagulation, combined contraceptives or hormonal replacement therapy, immobility, and undergoing major surgery). §In patients with difficult-to-treat rheumatoid arthritis, after two or more ineffective biological or targeted synthetic DMARDs (particularly TNF inhibitors), a biological or targeted synthetic DMARD with a different target (ie, mechanism of action) should be considered.

therapies. Other clinical trials have shown similar efficacy of combinations between methotrexate plus glucocorticoids versus methotrexate plus a biological DMARD.[59]

Glucocorticoids have a well-known risk profile of side-effects, especially if exposure to high doses is protracted.[60] In a multicentre randomised clinical trial (the GLORIA study), 451 patients with established rheumatoid arthritis and who were older than 65 years were randomly assigned to receive prednisolone 5 mg a day or placebo for 2 years.[61] Patients with rheumatoid arthritis who received prednisolone had better disease activity outcomes and showed less radiographic progression than those receiving placebo. In the prednisolone group, there was a higher incidence of side-effects compared with placebo, but the side-effects were primarily non-severe infections. However, because the cumulative dose of glucocorticoids has been mainly associated with increased mortality, long-term observation would be needed to discern the safety profile of glucocorticoids in this susceptible population (ie, older than 65 years and with active rheumatoid arthritis).

The position of the ACR and EULAR regarding the use of glucocorticoids in the management of rheumatoid arthritis is slightly divergent.[53,54] EULAR strongly recommends the use of short-term glucocorticoids as bridging therapy when initiating or changing DMARDs, with tapering and discontinuation performed as rapidly as is clinically feasible (ideally within 3 months), whereas the ACR cautions that glucocorticoids should not be systematically prescribed. The ACR, however, acknowledge the conditionality of this statement, given the frequent need to alleviate a patient's symptoms before the onset of DMARD action or in case of a flare. Intramuscular steroid is commonly used and is described as being effective for minimum dose with little rebound on steroid withdrawal, as suggested by EULAR.

The use of JAK inhibitors in patients with rheumatoid arthritis

Among the available treatment options for rheumatoid arthritis, JAK inhibitors represent the newest class of drugs. JAK inhibitors are small, targeted molecules that act through the inhibition of a family of protein kinases (eg, JAK 1, JAK 2, JAK 3, and tyrosine kinase 2), thus blocking the production of multiple cytokines implicated in the pathogenesis of rheumatoid arthritis, and have a different mechanism of action compared with biological DMARDs.[62] Unlike biological DMARDs, JAK inhibitors are orally administered. Known side-effects include an increase both in the reactivation of herpes zoster infection and in venous thromboembolism.[63,64] The efficacy, both in the improvement of clinical signs and symptoms and in the inhibition of progression of structural joint damage, and safety of JAK inhibitors in patients with rheumatoid arthritis have been documented in clinical trials.[63,64] In contrast to most biological DMARDs, JAK inhibitors have a higher proven efficacy than methotrexate monotherapy and have shown superiority to biological DMARDs when both are combined with methotrexate.[65–68] Both EULAR in 2019,[69] before the updated 2022 recommendations, and the latest ACR guidelines recommended that JAK inhibitors could be used as first-line advanced therapy in patients who do not reach the treatment target despite conventional synthetic DMARD therapy.

In the ORAL Surveillance randomised, open-label, post-authorisation, phase 3b–4 safety endpoint trial in 2022,[70] the incidence of major adverse cardiovascular events and cancer (excluding non-melanoma skin cancer) was compared between the JAK inhibitor tofacinitib (5 mg twice daily, which is the approved dose for rheumatoid arthritis, or 10 mg twice daily, as per treatment for ulcerative colitis) and TNF inhibitors adalimumab or etanercept in more than 4000 patients who were older than 50 years with active moderate-to-severe rheumatoid arthritis despite methotrexate treatment and who had one or more cardiovascular risk factors. In this study, major adverse cardiovascular events and cancer occurred more frequently in patients given tofacitinib than in patients given TNF inhibitors, with a hazard ratio of 1·33 (95% CI 0·91–1·94) for major adverse cardiovascular events and 1·48 (1·04–2·09) for cancers. In addition, during the trial, the investigators reduced the dose of tofacitinib from 10 mg twice daily to 5 mg twice daily after the observation of a higher incidence of pulmonary embolism and mortality in patients receiving tofacitinib 10 mg twice daily compared with those receiving a TNF inhibitor.

The results of ORAL Surveillance triggered an immediate response by the US Food and Drug Administration (FDA), who produced a warning (ie, a drug safety communication) regarding the increased risk of serious heart-related events, cancer, blood clots, and mortality for JAK inhibitors.[71] In this document, the FDA stated these drugs should be reserved for patients who have had an inadequate response or intolerance to at least one TNF inhibitor, thus downgrading JAK inhibitors to second line among advanced therapies. A similar document was subsequently published by the European Medicines Agency.[72]

In contrast to the previous recommendations in 2019, the new EULAR recommendations suggest considering the use of JAK inhibitors in patients without cardiovascular risk factors (eg, older than 65 years, previous or current smoking, diabetes, obesity, hypertension, and predisposing conditions for thromboembolic events) and previous or current malignancy other than successfully treated non-melanoma skin cancer.[54] Therefore, despite retaining the placement of JAK inhibitors as first-line therapy after conventional synthetic DMARDs in the latest EULAR recommendations, the safety profile will limit the use of JAK inhibitors to a specific population of patients younger than 65 years and without risk factors for cardiovascular events and malignancy.

Several important questions concerning the safety of JAK inhibitors and, therefore, their use in patients with rheumatoid arthritis are unanswered. Whether the risk profile, which emerged in the ORAL Surveillance trial,[70] would be applicable to more selective JAK inhibitors (ie, JAK1 or JAK 1/2) than tofacitinib (a pan-JAK inhibitor) or in patients with rheumatoid arthritis and without predisposing cardiovascular risk factors, needs to be established. Observational studies with large numbers of patients with rheumatoid arthritis have provided reassuring data on the safety profile of baricitinib and tofacitinib regarding the risk of major adverse cardiovascular events, venous thromboembolism, and malignancy (excluding non-melanoma skin cancer), when compared with TNF inhibitors.[73,74] A post-hoc analysis of the ORAL Surveillance study showed that patients with rheumatoid arthritis and without a history of atherosclerotic cardiovascular disease did not show an increased incidence of major adverse cardiovascular events when exposed to tofacitinib 5 mg twice daily compared with use of TNF inhibitors.[75]

In addition, in a post-hoc analysis of ORAL Surveillance, a subpopulation with a high risk of major adverse cardiovascular events and malignancies was identified based on age (ie, ≥65 years) and smoking (ie, ever smoked), with no statistically significant risk in patients without the risk factors, providing useful information on the risk assessment and decision making of treatment with tofacitinib in patients with rheumatoid arthritis.[76] In a nationwide cohort from Sweden including more than 27 000 patients with rheumatoid arthritis who were given JAK inhibitors, TNF inhibitors, or other biological DMARDs, therapy with an IL-6 receptor inhibitor showed a non-significant increase in venous thromboembolism risk compared with TNF inhibitors, especially in men.[77]

Remission in rheumatoid arthritis

The improvement in outcomes of patients with rheumatoid arthritis is reflected in various registries worldwide. An increasing number of patients with rheumatoid arthritis are now able to attain disease remission.[78,79]

Various composite indices are used in rheumatoid arthritis to define states of disease activity (eg, remission) and response to treatment. The most used is the Disease Activity Score, which uses a complex mathematical formula, with variables such as the number of tender and swollen joints out of 28, inflammatory markers (eg, C-reactive protein or erythrocyte sedimentation rate), and the patient assessment of global health status.[80,81] Other composite indices are the Clinical Disease Activity Index and the Simple Disease Activity Index for Rheumatoid Arthritis.[82,83] The ACR and EULAR have developed stringent remission criteria (ie, the ACR and EULAR Boolean criteria), in which the included variables (ie, tender joint count, swollen joint count, patient global assessment, and C-reactive protein) must all have a value of more than one (revised to more than two for patient global assessment).[84]

Both the ACR and EULAR considered the opportunity of tapering therapy in patients with sustained (ie, ≥6 months) clinical remission (EULAR) or low disease activity (ACR).[53,54] The ACR takes into account the high risk of flares that follows therapy discontinuation, with a potential negative effect on a patient's quality of life and

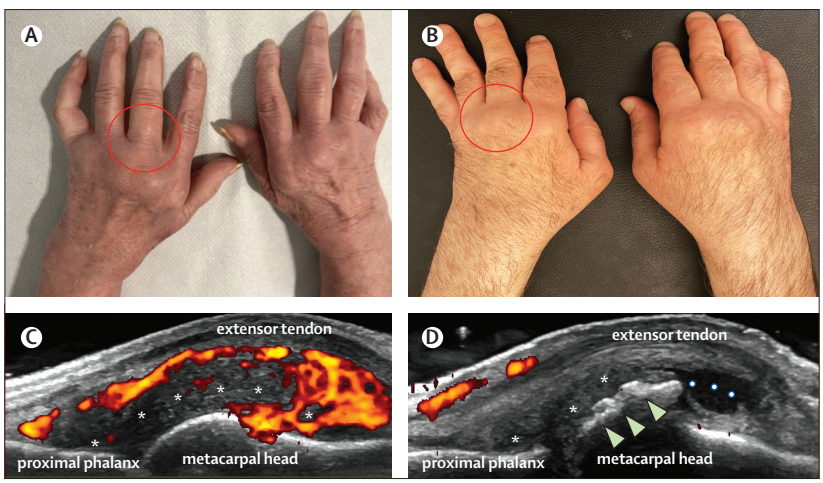


Figure 4: **Patients with difficult-to-treat rheumatoid arthritis and different sonographic scenarios**
Photographs of two patients with difficult-to-treat rheumatoid arthritis (A, B) and associated ultrasonography scenarios (C, D). Both patients had at least one ineffective conventional synthetic DMARD, at least two ineffective biological DMARDs with different mechanisms of action, a DAS-28 C-reactive protein of more than 3·2, and reduced quality of life related to rheumatoid arthritis disease activity at the time of the clinical visit. The patients were not on glucocorticoids. The metacarpophalangeal joints (A, B; red circles) were swollen and painful on clinical examination. In both patients, intra-articular synovial hypertrophy could be seen on ultrasonography (C, D; asterisks). Although there is evidence of diffuse power doppler signal (C; red areas) indicating active inflammation in one patient, the ultrasonography features of the second patient suggest advanced structural damage (D; arrowheads), mainly involving the metacarpal head (ie, joint subluxation and multiple bone erosions). In addition, no power doppler signal is detectable (D) but an area of joint effusion (D; white dots), together with the bony deformity involving the metacarpal head, could cause the clinical swelling. The presence of power doppler signal could indicate a persistent inflammatory process, which would be useful to identify persistent inflammatory refractory rheumatoid arthritis. Alternatively, chronic joint damage rather than active inflammation could explain the clinical signs and symptoms of the patient (D; non-inflammatory refractory rheumatoid arthritis). However, further studies are needed to define the prognostic value and implications of the different ultrsonography abnormalities in the population with difficult-to-treat rheumatoid arthritis. DAS-28=Disease Activity Score using 28 joints. DMARD=disease-modifying antirheumatic drug.

function, and conditionally recommend DMARD continuation at the current dose over therapy tapering.[85] If tapering is to be considered, both societies recommend a gradual tapering (ie, dose reduction or interval increase) over abrupt drug interruption. A large proportion of patients will have a flare when their DMARD treatment is tapered or discontinued.[86,87] This effect is mitigated by the fact that disease control is usually recaptured (in about 90% of patients) when treatment is resumed.[88]

The observation of frequent disease reactivation following therapy reduction or discontinuation has raised the hypothesis that reaching a threshold of disease remission according to the routinely used disease activity indices might not be sufficient to define a status of true disease remission. Some studies have shown the presence of subclinical synovitis on ultrasonography of patients in clinical remission, and this finding could arguably explain the radiographic progression observed in some patients with rheumatoid arthritis and no apparent clinically active disease.[89,90]

The concept of multidimension remission has emerged, which is based on the following levels of remission: (1) clinical (ie, absence of clinical signs of inflammation), (2) serological and imaging, and (3) immunological (ie, disappearance of autoantibodies).[91] Although reaching a clinical, serological, and imaging remission might represent an obtainable goal with rheumatoid arthritis therapeutic strategies, immunological remission has been observed infrequently in patients with rheumatoid arthritis, including those in sustained drug-free remission.[92] An alternative immunological remission based on the normalisation of T-cell subsets, proposed by Gul and colleagues, might be more practical.[93]

There is insufficient evidence that objective biomarkers can reliably predict the future risk of disease relapse.[94] A short disease duration, early remission induction with use of biological DMARDs, absence of radiographic damage, and low disease activity have been associated with successful therapy tapering in patients with rheumatoid arthritis in clinical remission.[95–97] However, abnormal T-cell subsets, ultrasonography subclinical inflammation, and inflammatory markers have been associated with disease relapse after therapy reduction or discontinuation.[98–100]

Challenges and future directions

Difficult-to-treat rheumatoid arthritis

One of the most challenging issues in rheumatoid arthritis is the management of patients with persistent active disease (including those with low disease activity), despite successive trials of DMARDs with varying mechanisms of action (ie, two or more biological or targeted synthetic DMARDs with different mechanisms of action, according to EULAR). The group of patients with multidrug-resistant rheumatoid arthritis, which has been recently termed difficult-to-treat rheumatoid arthritis by EULAR, represents an ongoing therapeutic challenge.[101,102]

The prevalence of difficult-to-treat rheumatoid arthritis reached 6% in a UK registry; however, the prevalence appears to be higher when conventional synthetic DMARDs, which are not accepted as criteria for difficult-to-treat rheumatoid arthritis in the EULAR definitions, are also considered.[103,104] In patients with rheumatoid arthritis, and especially in patients with difficult-to-treat rheumatoid arthritis, the correct identification of persistent active disease, and the distinction from joint pain and swelling from other causes, such as damage, osteoarthritis, or chronic non-articular pain, is crucial to guide pharmacological and non-pharmacological interventions. Continuous cycling of further DMARD therapy might be unnecessary in those without definitive evidence of an ongoing inflammatory joint process.

Two categories of patients with difficult-to-treat rheumatoid arthritis have been proposed: (1) patients with persistent inflammatory refractory rheumatoid arthritis (PIRRA), defined as the presence of unequivocal signs of ongoing inflammation (ie, ultrasonography inflammation or abnormal inflammatory markers); and (2) patients with rheumatoid arthritis with symptoms that persist despite the absence of discernible inflammation (non-inflammatory refractory rheumatoid arthritis

[NIRRA]; figure 4).[105] Despite this hypothesis, future investigations will have to clarify whether patients with PIRRA have different genetic or epigenetic mechanisms and immune pathways, which contribute to the immunopathogenesis of refractory inflammation compared with NIRRA and, therefore, have a different prognosis, long-term outcomes, and response to therapy. In addition, novel approaches, in which dual therapies, or bivalent antibodies, are used to simultaneously inhibit two different molecular pathways, are under consideration, but risk of infection might be a limiting factor.[106]

Precision medicine approach

In patients who do not respond to methotrexate, which occurs in up to 50% of people, and have poor prognostic factors, treatment escalation to biological or targeted synthetic DMARDs should be done. The answer to the question of which patients will respond to which type of treatment is unclear, as several meta-analysis and head-to-head clinical trials have not clearly shown a higher efficacy of any of the biological DMARDs (although some positive results have been published for targeted synthetic DMARDs) over others when combined with methotrexate.[107]

Therefore, the decision of which biological DMARD to use as first-line therapy is arbitrary and mainly based on cost. However, a few additional considerations include the patient's comorbidities and preferences, such as frequency or route of administration and, to a lesser extent, the presence of some biomarkers that have been inconsistently associated with a better response following specific therapy, such as positive rheumatoid arthritis autoantibodies for abatacept, rituximab, and TNF inhibitor responses, or increased C-reactive protein for IL-6 receptor inhibitor response.[108,109]

In patients who have started biological or targeted synthetic DMARD therapy, there are three possible outcomes, provided that the drug is well tolerated: (1) a good response, ideally disease remission; (2) no response or disease remission after 3–6 months of therapy (primary non-response); and (3) an initial good response, which is then followed by loss of response (secondary non-response).[110] In people who do not respond to one biological DMARD (ie, primary and secondary non-responders), there is a question whether an agent with a different mechanism of action versus a second agent with the same mode of action should be used. The scientific evidence is inconclusive, even though some studies have shown that use of a drug with the same mechanism of action can be associated with positive outcomes.[111,112] Drug immunogenicity (ie, the development of antidrug antibodies) has been regarded as one of the mechanisms responsible for secondary non-response (ie, loss of efficacy of a biological or targeted synthetic DMARD after an initial good response).[113] Notably, the response rate to biological DMARDs decreases with multiple unsuccessful drug treatments.[114]

Attention has turned to the investigation and characterisation of prominent cellular and molecular pathways and patterns within rheumatoid arthritis synovia across an array of patients through the use of ultrasonography-guided synovial biopsy techniques.[5] Several specific effector cell states in rheumatoid arthritis synovia have been identified as potential targets for therapy development, including MERTK^{+} macrophages, NOTCH3^{+} synovial fibroblasts, CD11c^{+} autoimmune-associated B cells, and PD-1^{+} peripheral helper T cells.[115–117] Moreover, in a clinical trial comparing an anti-CD20 antibody (rituximab) with an anti-IL-6 receptor inhibitor (tocilizumab), patients with a low or absent synovial B-cell molecular signature had lower responses with rituximab than tocilizumab.[118,119] Although still in early stages, these strategies offer hope that a precision medicine approach in rheumatoid arthritis, in which a DMARD is selected based on the synovial characteristics of the patient's tissue, could be forthcoming in the near future.

Prediction and prevention of rheumatoid arthritis

The development of clinical synovitis is widely accepted as the beginning of rheumatoid arthritis. Despite the considerable advances in the treatment and management of rheumatoid arthritis, once the disease develops, there is no cure for most patients. Long-term therapy is usually required, with important implications on the patient's quality of life and ability to work and for socioeconomic costs.

The detection of rheumatoid arthritis-related autoantibodies in the stored serum of patients with rheumatoid arthritis, particularly anti-CCP, years before the development of clinically evident disease, has been crucial for understanding preclinical rheumatoid arthritis.[120–122] The concept of a rheumatoid arthritis continuum has emerged. In the continuum, clinical synovitis is not the beginning of the disease, as per the traditional view, but the culmination of a series of mainly subclinical pathological events, such as autoimmunity and inflammation (figure 5).

Lessons from the early arthritis framework have shown that modifying the natural history of the disease and improving the patient's long-term outcome is possible. The hypothesis that the initiation of a treatment in the early stages of the rheumatoid arthritis continuum, before the development of clinically evident disease, might prevent, or at least delay, the onset of rheumatoid arthritis in individuals at high risk has been raised.[123,124]

In clinical trials on rheumatoid arthritis prevention, in which individuals at high risk are offered antirheumatic treatments, an accurate risk stratification is of paramount importance, especially to avoid the overtreatment of people who will never develop the disease. In the past decade, several research groups worldwide have built models for optimum risk prediction, which variably combine genetic, clinical, serological, and imaging risk

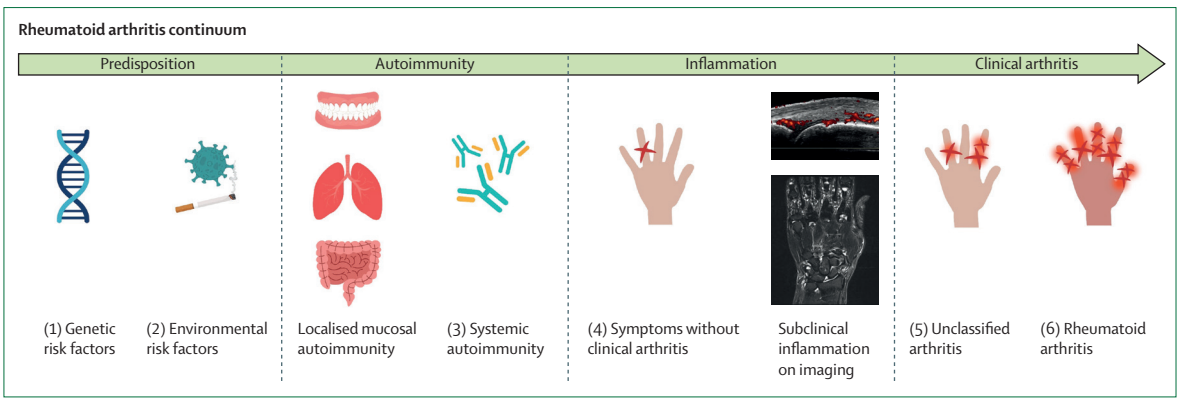


Figure 5: **The rheumatoid arthritis continuum: an overview of the rheumatoid arthritis preclinical phase**
Six stages along the rheumatoid arthritis continuum were defined in 2012 by the European Alliance of Associations for Rheumatology Committee on Investigative Rheumatology: (1) genetic risk factors for rheumatoid arthritis, (2) environmental risk factors for rheumatoid arthritis, (3) systemic autoimmunity associated with rheumatoid arthritis, (4) musculoskeletal symptoms without clinical arthritis, (5) unclassified arthritis, and (6) rheumatoid arthritis.[123] The phases do not necessarily occur in all patients (ie, seronegative rheumatoid arthritis) or in the same order in all patients. Increasing evidence suggests that systemic autoimmunity is preceded by a phase of mucosal dysbiosis with localised production of antibodies related to rheumatoid arthritis, mainly in the oral, lung, or gut mucosa (ie, localised mucosal autoimmunity). In addition, several studies have shown that a large proportion of patients with rheumatoid arthritis go through a stage of subclinical synovitis on imaging (eg, ultrasonography or MRI), which is, in people at high risk, associated with a substantial increase in the risk of progression to inflammatory arthritis (ie, subclinical inflammation on imaging).

factors for rheumatoid arthritis development.[125,126] A paper published in August, 2023, studied an extensive pre-rheumatoid arthritis population and provided scores for whom to refer from primary care and whom would be appropriate for intervention in secondary care.[127]

The most impactful intervention in rheumatoid arthritis would be its prevention. Multiple trials on rheumatoid arthritis prevention based on the use of drugs have been completed, and others are actively enrolling.[128] The ADJUST study showed that abatacept therapy for 6 months could delay the onset of rheumatoid arthritis in undifferentiated arthritis, while reducing anti-CCP levels and reducing erosions.[129]

In a randomised clinical trial, 83 patients with arthralgia and positive anti-CCP antibody or RF received two intramuscular injections of 100 mg dexamethasone at baseline and at 6 weeks or placebo.[130] Patients receiving dexamethasone showed a significant and sustained reduction of autoantibodies, which was the primary endpoint, compared with the placebo group. However, the rate of progression to rheumatoid arthritis was similar in the two groups (dexamethasone 20% *vs* placebo 21%) over 26 months.

In the PRAIRI trial, 81 individuals with positive anti-CCP antibodies and RF with MRI or ultrasonography evidence of inflammation or increased C-reactive protein were randomly assigned to receive a single dose of rituximab (1000 mg with 100 mg methylprednisolone premedication) or placebo.[131] In individuals receiving rituximab, there was a significant delay in the development of rheumatoid arthritis compared with the placebo group. 16 (40%) participants of 40 developed arthritis after a median of 11.5 months (IQR 2·5–15·0, range 1·0–40·0 months) in the placebo group and 14 (34%) participants of 41 developed arthritis after a median of 16·5 months (IQR 9·0–28·0, range 1·0–37·0 months) in the rituximab group. However, the rates of rheumatoid arthritis development after a follow-up of 29 months did not differ between the two groups (rituximab 34% *vs* placebo 40%).

In a randomised clinical trial, 236 individuals with clinically suspect arthralgia and MRI-detected subclinical joint inflammation were randomly assigned to 12 months therapy of oral methotrexate (up to 25 mg/week) plus a single intramuscular steroid injection (n=119), or placebo (n=117).[132] The trial did not show an ability of methotrexate versus placebo to reduce the rate of progression to rheumatoid arthritis after 2 years, although improvement in symptoms and MRI features was superior in the methotrexate group.

Other studies have shown the potential ability of abatacept to suppress subclinical synovitis on MRI scans in ACPA-positive individuals at high risk with MRI signs of inflammation.[133] The preliminary results of the Arthritis Prevention in the Preclinical Phase of Rheumatoid Arthritis with Abatacept trial showed a reduction in the development of rheumatoid arthritis over 2 years in individuals at high risk given abatacept.[134] Alternatively, an interim analysis of the Stop Rheumatoid Arthritis trial, which included 144 CCP3-positive individuals with or without arthralgia and imaging inflammation, did not show the ability of hydroxychloroquine to prevent the development of inflammatory arthritis compared with placebo; therefore, the trial was stopped before the expected conclusion.[135]

A potential interpretation for these studies is that an earlier stage in the rheumatoid arthritis continuum (ie, before the development of subclinical joint disease) might be the correct target for prevention.[136,137] Alternatively, not all patients with arthralgia and

Search strategy and selection criteria

The terms "rheumatoid arthritis" combined with "epidemiology", "pathogenesis", "clinical manifestations", "diagnosis", "classification", "management", "therapy", "prediction", and "prevention" were searched on MEDLINE from Jan 1, 2000 to May 11, 2023 with no language restrictions. References of selected articles were considered for the identification of relevant data. American College of Rheumatology and European Alliance of Associations for Rheumatology guidelines and recommendations for the management of rheumatoid arthritis (and systematic literature reviews relevant to this topic) were reviewed. We prioritised articles that were published in the past 5 years but also included impactful older publications, including those that preceded the start date of the search.

subclinical joint disease progress to rheumatoid arthritis[138] and patients who progress to rheumatoid arthritis have other risk factors present, such as high levels of antibodies or increased inflammatory markers.[139] The pivotal mechanisms implicated in progressing from subclinical to clinical disease are poorly understood, and targets for interventions have not been definitively identified. Preclinical rheumatoid arthritis, however defined, has considerable morbidity that responds to treatment, and guidelines for management are in development.

Conclusions

The management and outcomes of patients with rheumatoid arthritis have drastically improved in the last two decades. The recommendations for managing rheumatoid arthritis are changing rapidly, due to newly developed therapies and evolving scientific evidence. In this Seminar, the new ACR and EULAR recommendations are discussed, the controversies regarding the use of glucocorticoids and the uncertainty around JAK inhibitors have been highlighted, and the approaches to tapering therapy in patients who are in remission have been outlined. There is a clear need for reliable biomarkers for diagnosis, prognosis, and especially for response to therapy due to a proportion of patients not responding to multiple biological or targeted synthetic DMARDs. New areas of research have been illustrated, including difficult-to-treat rheumatoid arthritis, precision medicine, and attempts to delay or prevent arthritis.

Contributors

All authors discussed and agreed on the structure of the Seminar and contributed equally to the search of the papers. ADM prepared the images and tables and wrote the first draft of the manuscript, which was critically reviewed by JMB and PE. All authors have seen and approved the final version of the manuscript.

Declaration of interests

PE has received grants from Lilly and Samsung; consulting fees from BMS, Boehringer-Ingelheim, Lilly, and Novartis; payment and honoraria from Galapagos, Lilly, and Novartis; support for attending meetings or travel from Lilly; and has participated on a data safety monitoring board or advisory board with AstraZeneca. All other authors declare no competing interests.

References

1 Aletaha D, Smolen JS. Diagnosis and management of rheumatoid arthritis: a review. *JAMA* 2018; **320:** 1360–72.
2 Smolen JS, Aletaha D, McInnes IB. Rheumatoid arthritis. *Lancet* 2016; **388:** 2023–38.
3 Finckh A, Gilbert B, Hodkinson B, et al. Global epidemiology of rheumatoid arthritis. *Nat Rev Rheumatol* 2022; **18:** 591–602.
4 Scott IC, Whittle R, Bailey J, et al. Rheumatoid arthritis, psoriatic arthritis, and axial spondyloarthritis epidemiology in England from 2004 to 2020: an observational study using primary care electronic health record data. *Lancet Reg Health Eur* 2022; **23:** 100519.
5 Gravallese EM, Firestein GS. Rheumatoid arthritis—common origins, divergent mechanisms. *N Engl J Med* 2023; **388:** 529–42.
6 Alivernini S, Firestein GS, McInnes IB. The pathogenesis of rheumatoid arthritis. *Immunity* 2022; **55:** 2255–70.
7 Hemminki K, Li X, Sundquist J, Sundquist K. Familial associations of rheumatoid arthritis with autoimmune diseases and related conditions. *Arthritis Rheum* 2009; **60:** 661–68.
8 Gregersen PK, Silver J, Winchester RJ. The shared epitope hypothesis—an approach to understanding the molecular genetics of susceptibility to rheumatoid arthritis. *Arthritis Rheum* 1987; **30:** 1205–13.
9 Wouters F, Maurits MP, van Boheemen L, et al. Determining in which pre-arthritis stage HLA-shared epitope alleles and smoking exert their effect on the development of rheumatoid arthritis. *Ann Rheum Dis* 2022; **81:** 48–55.
10 Karlson EW, Chibnik LB, Cui J, et al. Associations between human leukocyte antigen, PTPN22, CTLA4 genotypes and rheumatoid arthritis phenotypes of autoantibody status, age at diagnosis and erosions in a large cohort study. *Ann Rheum Dis* 2008; **67:** 358–63.
11 Plenge RM, Padyukov L, Remmers EF, et al. Replication of putative candidate-gene associations with rheumatoid arthritis in >4000 samples from North America and Sweden: association of susceptibility with PTPN22, CTLA4, and PADI4. *Am J Hum Genet* 2005; **77:** 1044–60.
12 Ishigaki K, Sakaue S, Terao C, et al. Multi-ancestry genome-wide association analyses identify novel genetic mechanisms in rheumatoid arthritis. *Nat Genet* 2022; **54:** 1640–51.
13 Nemtsova MV, Zaletaev DV, Bure IV, et al. Epigenetic changes in the pathogenesis of rheumatoid arthritis. *Front Genet* 2019; **10:** 570.
14 Conrad N, Misra S, Verbakel JY, et al. Incidence, prevalence, and co-occurrence of autoimmune disorders over time and by age, sex, and socioeconomic status: a population-based cohort study of 22 million individuals in the UK. *Lancet* 2023; **401:** 1878–90.
15 Sugiyama D, Nishimura K, Tamaki K, et al. Impact of smoking as a risk factor for developing rheumatoid arthritis: a meta-analysis of observational studies. *Ann Rheum Dis* 2010; **69:** 70–81.
16 Deane KD, Demoruelle MK, Kelmenson LB, Kuhn KA, Norris JM, Holers VM. Genetic and environmental risk factors for rheumatoid arthritis. *Best Pract Res Clin Rheumatol* 2017; **31:** 3–18.
17 Di Giuseppe D, Alfredsson L, Bottai M, Askling J, Wolk A. Long term alcohol intake and risk of rheumatoid arthritis in women: a population based cohort study. *BMJ* 2012; **345:** e4230.
18 Myasoedova E, Davis J, Matteson EL, Crowson CS. Is the epidemiology of rheumatoid arthritis changing? Results from a population-based incidence study, 1985–2014. *Ann Rheum Dis* 2020; **79:** 440–44.
19 van Venrooij WJ, van Beers JJ, Pruijn GJ. Anti-CCP antibodies: the past, the present and the future. *Nat Rev Rheumatol* 2011; **7:** 391–98.
20 Bas S, Genevay S, Meyer O, Gabay C. Anti-cyclic citrullinated peptide antibodies, IgM and IgA rheumatoid factors in the diagnosis and prognosis of rheumatoid arthritis. *Rheumatology (Oxford)* 2003; **42:** 677–80.
21 Shi J, van de Stadt LA, Levarht EW, et al. Anti-carbamylated protein (anti-CarP) antibodies precede the onset of rheumatoid arthritis. *Ann Rheum Dis* 2014; **73:** 780–83.
22 Shi J, van Steenbergen HW, van Nies JA, et al. The specificity of anti-carbamylated protein antibodies for rheumatoid arthritis in a setting of early arthritis. *Arthritis Res Ther* 2015; **17:** 339.

23 Mankia K, Emery P. Preclinical rheumatoid arthritis: progress toward prevention. *Arthritis Rheumatol* 2016; **68:** 779–88.

24 Zhang X, Zhang D, Jia H, et al. The oral and gut microbiomes are perturbed in rheumatoid arthritis and partly normalized after treatment. *Nat Med* 2015; **21:** 895–905.

25 Holers VM, Demoruelle MK, Kuhn KA, et al. Rheumatoid arthritis and the mucosal origins hypothesis: protection turns to destruction. *Nat Rev Rheumatol* 2018; **14:** 542–57.

26 Nishimura K, Sugiyama D, Kogata Y, et al. Meta-analysis: diagnostic accuracy of anti-cyclic citrullinated peptide antibody and rheumatoid factor for rheumatoid arthritis. *Ann Intern Med* 2007; **146:** 797–808.

27 van den Broek M, Dirven L, Klarenbeek NB, et al. The association of treatment response and joint damage with ACPA-status in recent-onset RA: a subanalysis of the 8-year follow-up of the BeSt study. *Ann Rheum Dis* 2012; **71:** 245–48.

28 McInnes IB, Schett G. The pathogenesis of rheumatoid arthritis. *N Engl J Med* 2011; **365:** 2205–19.

29 Choy E. Understanding the dynamics: pathways involved in the pathogenesis of rheumatoid arthritis. *Rheumatology (Oxford)* 2012; **51** (suppl 5): v3–11.

30 Guo Q, Wang Y, Xu D, Nossent J, Pavlos NJ, Xu J. Rheumatoid arthritis: pathological mechanisms and modern pharmacologic therapies. *Bone Res* 2018; **6:** 15.

31 Feldmann M, Maini SR. Role of cytokines in rheumatoid arthritis: an education in pathophysiology and therapeutics. *Immunol Rev* 2008; **223:** 7–19.

32 Smolen JS, Aletaha D, Barton A, et al. Rheumatoid arthritis. *Nat Rev Dis Primers* 2018; **4:** 18001.

33 Mankia K, D'Agostino MA, Rowbotham E, et al. MRI inflammation of the hand interosseous tendons occurs in anti-CCP-positive at-risk individuals and may precede the development of clinical synovitis. *Ann Rheum Dis* 2019; **78:** 781–86.

34 Turesson C, O'Fallon WM, Crowson CS, Gabriel SE, Matteson EL. Extra-articular disease manifestations in rheumatoid arthritis: incidence trends and risk factors over 46 years. *Ann Rheum Dis* 2003; **62:** 722–27.

35 Turesson C. Extra-articular rheumatoid arthritis. *Curr Opin Rheumatol* 2013; **25:** 360–66.

36 Aletaha D, Neogi T, Silman AJ, et al. 2010 Rheumatoid arthritis classification criteria: an American College of Rheumatology/ European League Against Rheumatism collaborative initiative. *Arthritis Rheum* 2010; **62:** 2569–81.

37 Arnett FC, Edworthy SM, Bloch DA, et al. The American Rheumatism Association 1987 revised criteria for the classification of rheumatoid arthritis. *Arthritis Rheum* 1988; **31:** 315–24.

38 Radner H, Neogi T, Smolen JS, Aletaha D. Performance of the 2010 ACR/EULAR classification criteria for rheumatoid arthritis: a systematic literature review. *Ann Rheum Dis* 2014; **73:** 114–23.

39 Di Matteo A, Mankia K, Azukizawa M, Wakefield RJ. The role of musculoskeletal ultrasound in the rheumatoid arthritis continuum. *Curr Rheumatol Rep* 2020; **22:** 41.

40 Schett G, Gravallese E. Bone erosion in rheumatoid arthritis: mechanisms, diagnosis and treatment. *Nat Rev Rheumatol* 2012; **8:** 656–64.

41 Colebatch AN, Edwards CJ, Østergaard M, et al. EULAR recommendations for the use of imaging of the joints in the clinical management of rheumatoid arthritis. *Ann Rheum Dis* 2013; **72:** 804–14.

42 Baillet A, Gaujoux-Viala C, Mouterde G, et al. Comparison of the efficacy of sonography, magnetic resonance imaging and conventional radiography for the detection of bone erosions in rheumatoid arthritis patients: a systematic review and meta-analysis. *Rheumatology (Oxford)* 2011; **50:** 1137–47.

43 Haavardsholm EA, Aga A-B, Olsen IC, et al. Ultrasound in management of rheumatoid arthritis: ARCTIC randomised controlled strategy trial. *BMJ* 2016; **354:** i4205.

44 Dale J, Stirling A, Zhang R, et al. Targeting ultrasound remission in early rheumatoid arthritis: the results of the TaSER study, a randomised clinical trial. *Ann Rheum Dis* 2016; **75:** 1043–50.

45 Smolen JS, Aletaha D, Bijlsma JW, et al. Treating rheumatoid arthritis to target: recommendations of an international task force. *Ann Rheum Dis* 2010; **69:** 631–37.

46 Burgers LE, Raza K, van der Helm-van Mil AH. Window of opportunity in rheumatoid arthritis—definitions and supporting evidence: from old to new perspectives. *RMD Open* 2019; **5:** e000870.

47 Pincus T, Yazici Y, Sokka T, Aletaha D, Smolen JS. Methotrexate as the "anchor drug" for the treatment of early rheumatoid arthritis. *Clin Exp Rheumatol* 2003; **21** (suppl 31)**:** S179–85.

48 Frisell T, Bower H, Morin M, et al. Safety of biological and targeted synthetic disease-modifying antirheumatic drugs for rheumatoid arthritis as used in clinical practice: results from the ARTIS programme. *Ann Rheum Dis* 2023; **82:** 601–10.

49 Kerschbaumer A, Sepriano A, Bergstra SA, et al. Efficacy of synthetic and biological DMARDs: a systematic literature review informing the 2022 update of the EULAR recommendations for the management of rheumatoid arthritis. *Ann Rheum Dis* 2023; **82:** 95–106.

50 Nam JL, Ramiro S, Gaujoux-Viala C, et al. Efficacy of biological disease-modifying antirheumatic drugs: a systematic literature review informing the 2013 update of the EULAR recommendations for the management of rheumatoid arthritis. *Ann Rheum Dis* 2014; **73:** 516–28.

51 Jani M, Barton A, Warren RB, Griffiths CE, Chinoy H. The role of DMARDs in reducing the immunogenicity of TNF inhibitors in chronic inflammatory diseases. *Rheumatology (Oxford)* 2014; **53:** 213–22.

52 Emery P, Pope JE, Kruger K, et al. Efficacy of monotherapy with biologics and JAK inhibitors for the treatment of rheumatoid arthritis: a systematic review. *Adv Ther* 2018; **35:** 1535–63.

53 Fraenkel L, Bathon JM, England BR, et al. 2021 American College of Rheumatology guideline for the treatment of rheumatoid arthritis. *Arthritis Rheumatol* 2021; **73:** 1108–23.

54 Smolen JS, Landewé RBM, Bergstra SA, et al. EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2022 update. *Ann Rheum Dis* 2023; **82:** 3–18.

55 Doumen M, Pazmino S, Bertrand D, Westhovens R, Verschueren P. Glucocorticoids in rheumatoid arthritis: balancing benefits and harm by leveraging the therapeutic window of opportunity. *Joint Bone Spine* 2023; **90:** 105491.

56 Graudal N, Jürgens G. Similar effects of disease-modifying antirheumatic drugs, glucocorticoids, and biologic agents on radiographic progression in rheumatoid arthritis: meta-analysis of 70 randomized placebo-controlled or drug-controlled studies, including 112 comparisons. *Arthritis Rheum* 2010; **62:** 2852–63.

57 Wassenberg S, Rau R, Steinfeld P, Zeidler H. Very low-dose prednisolone in early rheumatoid arthritis retards radiographic progression over two years: a multicenter, double-blind, placebo-controlled trial. *Arthritis Rheum* 2005; **52:** 3371–80.

58 Hetland ML, Haavardsholm EA, Rudin A, et al. Active conventional treatment and three different biological treatments in early rheumatoid arthritis: phase 4 investigator initiated, randomised, observer blinded clinical trial. *BMJ* 2020; **371:** m4328.

59 Kerschbaumer A, Sepriano A, Smolen JS, et al. Efficacy of pharmacological treatment in rheumatoid arthritis: a systematic literature research informing the 2019 update of the EULAR recommendations for management of rheumatoid arthritis. *Ann Rheum Dis* 2020; **79:** 744–59.

60 Hua C, Buttgereit F, Combe B. Glucocorticoids in rheumatoid arthritis: current status and future studies. *RMD Open* 2020; **6:** e000536.

61 Boers M, Hartman L, Opris-Belinski D, et al. Low dose, add-on prednisolone in patients with rheumatoid arthritis aged 65+: the pragmatic randomised, double-blind placebo-controlled GLORIA trial. *Ann Rheum Dis* 2022; **81:** 925–36.

62 Nash P, Kerschbaumer A, Dörner T, et al. Points to consider for the treatment of immune-mediated inflammatory diseases with Janus kinase inhibitors: a consensus statement. *Ann Rheum Dis* 2021; **80:** 71–87.

63 Wang F, Sun L, Wang S, et al. Efficacy and safety of tofacitinib, baricitinib, and upadacitinib for rheumatoid arthritis: a systematic review and meta-analysis. *Mayo Clin Proc* 2020; **95:** 1404–19.

64 Kerschbaumer A, Smolen JS, Nash P, et al. Points to consider for the treatment of immune-mediated inflammatory diseases with Janus kinase inhibitors: a systematic literature research. *RMD Open* 2020; **6:** e001374.

65 Lee EB, Fleischmann R, Hall S, et al. Tofacitinib versus methotrexate in rheumatoid arthritis. *N Engl J Med* 2014; **370:** 2377–86.

66 van Vollenhoven R, Takeuchi T, Pangan AL, et al. Efficacy and safety of upadacitinib monotherapy in methotrexate-naive patients with moderately-to-severely active rheumatoid arthritis (SELECT-EARLY): a multicenter, multi-country, randomized, double-blind, active comparator-controlled trial. *Arthritis Rheumatol* 2020; **72:** 1607–20.

67 Fleischmann R, Schiff M, van der Heijde D, et al. Baricitinib, methotrexate, or combination in patients with rheumatoid arthritis and no or limited prior disease-modifying antirheumatic drug treatment. *Arthritis Rheumatol* 2017; **69:** 506–17.

68 Taylor PC, Keystone EC, van der Heijde D, et al. Baricitinib versus placebo or adalimumab in rheumatoid arthritis. *N Engl J Med* 2017; **376:** 652–62.

69 Smolen JS, Landewé RBM, Bijlsma JWJ, et al. EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2019 update. *Ann Rheum Dis* 2020; **79:** 685–99.

70 Ytterberg SR, Bhatt DL, Mikuls TR, et al. Cardiovascular and cancer risk with tofacitinib in rheumatoid arthritis. *N Engl J Med* 2022; **386:** 316–26.

71 Food and Drug Administration. FDA requires warnings about increased risk of serious heart-related events, cancer, blood clots, and death for JAK inhibitors that treat certain chronic inflammatory conditions. Dec 7, 2021. https://www.fda.gov/drugs/drug-safety-and-availability/fda-requires-warnings-about-increased-risk-serious-heart-related-events-cancer-blood-clots-and-death (accessed May 11, 2023).

72 European Medicines Agency. Xeljanz (tofacitinib): increased risk of major adverse cardiovascular events and malignancies with use of tofacitinib relative to TNF-alpha inhibitors. July 6, 2021. https://www.ema.europa.eu/en/medicines/dhpc/xeljanz-tofacitinib-increased-risk-major-adverse-cardiovascular-events-malignancies-use-tofacitinib (accessed May 11, 2023).

73 Huss V, Bower H, Hellgren K, Frisell T, Askling J. Cancer risks with JAKi and biological disease-modifying antirheumatic drugs in patients with rheumatoid arthritis or psoriatic arthritis: a national real-world cohort study. *Ann Rheum Dis* 2023; **82:** 911–19.

74 Hoisnard L, Pina Vegas L, Dray-Spira R, Weill A, Zureik M, Sbidian E. Risk of major adverse cardiovascular and venous thromboembolism events in patients with rheumatoid arthritis exposed to JAK inhibitors versus adalimumab: a nationwide cohort study. *Ann Rheum Dis* 2023; **82:** 182–88.

75 Charles-Schoeman C, Buch MH, Dougados M, et al. Risk of major adverse cardiovascular events with tofacitinib versus tumour necrosis factor inhibitors in patients with rheumatoid arthritis with or without a history of atherosclerotic cardiovascular disease: a post hoc analysis from ORAL Surveillance. *Ann Rheum Dis* 2023; **82:** 119–29.

76 Kristensen LE, Danese S, Yndestad A, et al. Identification of two tofacitinib subpopulations with different relative risk versus TNF inhibitors: an analysis of the open label, randomised controlled study ORAL Surveillance. *Ann Rheum Dis* 2023; **82:** 901–10.

77 Molander V, Bower H, Frisell T, Delcoigne B, Di Giuseppe D, Askling J. Venous thromboembolism with JAK inhibitors and other immune-modulatory drugs: a Swedish comparative safety study among patients with rheumatoid arthritis. *Ann Rheum Dis* 2023; **82:** 189–97.

78 Einarsson JT, Willim M, Saxne T, Geborek P, Kapetanovic MC. Secular trends of sustained remission in rheumatoid arthritis, a nationwide study in Sweden. *Rheumatology (Oxford)* 2020; **59:** 205–12.

79 Aga AB, Lie E, Uhlig T, et al. Time trends in disease activity, response and remission rates in rheumatoid arthritis during the past decade: results from the NOR-DMARD study 2000–10. *Ann Rheum Dis* 2015; **74:** 381–88.

80 van der Heijde DMFM, van 't Hof MA, van Riel PLCM, et al. Judging disease activity in clinical practice in rheumatoid arthritis: first step in the development of a disease activity score. *Ann Rheum Dis* 1990; **49:** 916–20.

81 Prevoo MLL, van 't Hof MA, Kuper HH, van Leeuwen MA, van de Putte LB, van Riel PL. Modified disease activity scores that include twenty-eight-joint counts—development and validation in a prospective longitudinal study of patients with rheumatoid arthritis. *Arthritis Rheum* 1995; **38:** 44–48.

82 Aletaha D, Nell VPK, Stamm T, et al. Acute phase reactants add little to composite disease activity indices for rheumatoid arthritis: validation of a clinical activity score. *Arthritis Res Ther* 2005; **7:** R796–806.

83 Smolen JS, Breedveld FC, Schiff MH, et al. A simplified disease activity index for rheumatoid arthritis for use in clinical practice. *Rheumatology (Oxford)* 2003; **42:** 244–57.

84 Felson DT, Smolen JS, Wells G, et al. American College of Rheumatology/European League against Rheumatism provisional definition of remission in rheumatoid arthritis for clinical trials. *Ann Rheum Dis* 2011; **70:** 404–13.

85 Smolen JS, Pedersen R, Jones H, Mahgoub E, Marshall L. Impact of flare on radiographic progression after etanercept continuation, tapering or withdrawal in patients with rheumatoid arthritis. *Rheumatology (Oxford)* 2020; **59:** 153–64.

86 Terslev L, Ostergaard M, Georgiadis S, et al. Flare during tapering of biological DMARDs in patients with rheumatoid arthritis in routine care: characteristics and predictors. *RMD Open* 2022; **8:** e002796.

87 Amstad A, Papagiannoulis E, Scherer A, et al. Comparison of drug retention of TNF inhibitors, other biologics and JAK inhibitors in RA patients who discontinued JAK inhibitor therapy. *Rheumatology (Oxford)* 2022; **62:** 89–97.

88 Schett G, Tanaka Y, Isaacs JD. Why remission is not enough: underlying disease mechanisms in RA that prevent cure. *Nat Rev Rheumatol* 2021; **17:** 135–44.

89 Brown AK, Quinn MA, Karim Z, et al. Presence of significant synovitis in rheumatoid arthritis patients with disease-modifying antirheumatic drug-induced clinical remission: evidence from an imaging study may explain structural progression. *Arthritis Rheum* 2006; **54:** 3761–73.

90 Brown AK, Conaghan PG, Karim Z, et al. An explanation for the apparent dissociation between clinical remission and continued structural deterioration in rheumatoid arthritis. *Arthritis Rheum* 2008; **58:** 2958–67.

91 Schett G, Emery P, Tanaka Y, et al. Tapering biologic and conventional DMARD therapy in rheumatoid arthritis: current evidence and future directions. *Ann Rheum Dis* 2016; **75:** 1428–37.

92 Boeters DM, Burgers LE, Toes RE, van der Helm-van Mil A. Does immunological remission, defined as disappearance of autoantibodies, occur with current treatment strategies? A long-term follow-up study in rheumatoid arthritis patients who achieved sustained DMARD-free status. *Ann Rheum Dis* 2019; **78:** 1497–504.

93 Gul HL, Eugenio G, Rabin T, et al. Defining remission in rheumatoid arthritis: does it matter to the patient? A comparison of multi-dimensional remission criteria and patient reported outcomes. *Rheumatology (Oxford)* 2020; **59:** 613–21.

94 Schlager L, Loiskandl M, Aletaha D, Radner H. Predictors of successful discontinuation of biologic and targeted synthetic DMARDs in patients with rheumatoid arthritis in remission or low disease activity: a systematic literature review. *Rheumatology (Oxford)* 2020; **59:** 324–34.

95 Huizinga TW, Conaghan PG, Martin-Mola E, et al. Clinical and radiographic outcomes at 2 years and the effect of tocilizumab discontinuation following sustained remission in the second and third year of the ACT-RAY study. *Ann Rheum Dis* 2015; **74:** 35–43.

96 Ghiti Moghadam M, Vonkeman HE, Ten Klooster PM, et al. Stopping tumor necrosis factor inhibitor treatment in patients with established rheumatoid arthritis in remission or with stable low disease activity: a pragmatic multicenter, open-label randomized controlled trial. *Arthritis Rheumatol* 2016; **68:** 1810–17.

97 Bijlsma JWJ, Welsing PMJ, Woodworth TG, et al. Early rheumatoid arthritis treated with tocilizumab, methotrexate, or their combination (U-Act-Early): a multicentre, randomised, double-blind, double-dummy, strategy trial. *Lancet* 2016; **388:** 343–55.

98 Filippou G, Sakellariou G, Scirè CA, et al. The predictive role of ultrasound-detected tenosynovitis and joint synovitis for flare in patients with rheumatoid arthritis in stable remission—results of an Italian multicentre study of the Italian Society for Rheumatology Group for Ultrasound: the STARTER study. *Ann Rheum Dis* 2018; **77:** 1283–89.

99 Gul HL, Di Matteo A, Mankia K, Wu J, Ponchel F, Emery P. Can biomarkers predict successful tapering of conventional disease-modifying therapy in rheumatoid arthritis patients in stable remission? *Clin Exp Rheumatol* 2023; **41:** 126–36.

100 Burgoyne CH, Field SL, Brown AK, et al. Abnormal T cell differentiation persists in patients with rheumatoid arthritis in clinical remission and predicts relapse. *Ann Rheum Dis* 2008; **67:** 750–57.

101 Nagy G, Roodenrijs NMT, Welsing PM, et al. EULAR definition of difficult-to-treat rheumatoid arthritis. *Ann Rheum Dis* 2021; **80:** 31–35.

102 Nagy G, Roodenrijs NMT, Welsing PMJ, et al. EULAR points to consider for the management of difficult-to-treat rheumatoid arthritis. *Ann Rheum Dis* 2022; **81:** 20–33.

103 Kearsley-Fleet L, Davies R, De Cock D, et al. Biologic refractory disease in rheumatoid arthritis: results from the British Society for Rheumatology Biologics Register for Rheumatoid Arthritis. *Ann Rheum Dis* 2018; **77:** 1405–12.

104 Bécède M, Alasti F, Gessl I, et al. Risk profiling for a refractory course of rheumatoid arthritis. *Semin Arthritis Rheum* 2019; **49:** 211–17.

105 Buch MH, Eyre S, McGonagle D. Persistent inflammatory and non-inflammatory mechanisms in refractory rheumatoid arthritis. *Nat Rev Rheumatol* 2021; **17:** 17–33.

106 Boleto G, Kanagaratnam L, Dramé M, Salmon JH. Safety of combination therapy with two bDMARDs in patients with rheumatoid arthritis: a systematic review and meta-analysis. *Semin Arthritis Rheum* 2019; **49:** 35–42.

107 Fleischmann R, Landewé R, Smolen JS. Review of head-to-head study designs in rheumatoid arthritis. *Semin Arthritis Rheum* 2016; **46:** 279–85.

108 Isaacs JD, Cohen SB, Emery P, et al. Effect of baseline rheumatoid factor and anticitrullinated peptide antibody serotype on rituximab clinical response: a meta-analysis. *Ann Rheum Dis* 2013; **72:** 329–36.

109 Wang J, Devenport J, Low JM, Yu D, Hitraya E. Relationship between baseline and early changes in C-reactive protein and interleukin-6 levels and clinical response to tocilizumab in rheumatoid arthritis. *Arthritis Care Res (Hoboken)* 2016; **68:** 882–85.

110 Vallejo-Yagüe E, Keystone EC, Kandhasamy S, Micheroli R, Finckh A, Burden AM. Primary and secondary non-response: in need of operational definitions in observational studies. *Ann Rheum Dis* 2021; **80:** 961–64.

111 Smolen JS, Kay J, Doyle MK, et al. Golimumab in patients with active rheumatoid arthritis after treatment with tumour necrosis factor alpha inhibitors (GO-AFTER study): a multicentre, randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2009; **374:** 210–21.

112 Emery P, van Hoogstraten H, Thangavelu K, Mangan E, St John G, Verschueren P. Subcutaneous sarilumab in patients with rheumatoid arthritis who previously received subcutaneous sarilumab or intravenous tocilizumab: an open-label extension of a randomized clinical trial. *ACR Open Rheumatol* 2020; **2:** 672–80.

113 Moots RJ, Xavier RM, Mok CC, et al. The impact of anti-drug antibodies on drug concentrations and clinical outcomes in rheumatoid arthritis patients treated with adalimumab, etanercept, or infliximab: results from a multinational, real-world clinical practice, non-interventional study. *PLoS One* 2017; **12:** e0175207.

114 Aletaha D, Maa JF, Chen S, et al. Effect of disease duration and prior disease-modifying antirheumatic drug use on treatment outcomes in patients with rheumatoid arthritis. *Ann Rheum Dis* 2019; **78:** 1609–15.

115 Alivernini S, MacDonald L, Elmesmari A, et al. Distinct synovial tissue macrophage subsets regulate inflammation and remission in rheumatoid arthritis. *Nat Med* 2020; **26:** 1295–306.

116 Wei K, Korsunsky I, Marshall JL, et al. Notch signalling drives synovial fibroblast identity and arthritis pathology. *Nature* 2020; **582:** 259–64.

117 Zhang F, Wei K, Slowikowski K, et al. Defining inflammatory cell states in rheumatoid arthritis joint synovial tissues by integrating single-cell transcriptomics and mass cytometry. *Nat Immunol* 2019; **20:** 928–42.

118 Humby F, Durez P, Buch MH, et al. Rituximab versus tocilizumab in anti-TNF inadequate responder patients with rheumatoid arthritis (R4RA): 16-week outcomes of a stratified, biopsy-driven, multicentre, open-label, phase 4 randomised controlled trial. *Lancet* 2021; **397:** 305–17.

119 Rivellese F, Surace AEA, Goldmann K, et al. Rituximab versus tocilizumab in rheumatoid arthritis: synovial biopsy-based biomarker analysis of the phase 4 R4RA randomized trial. *Nat Med* 2022; **28:** 1256–68.

120 del Puente A, Knowler WC, Pettitt DJ, Bennett PH. The incidence of rheumatoid arthritis is predicted by rheumatoid factor titer in a longitudinal population study. *Arthritis Rheum* 1988; **31:** 1239–44.

121 Rantapää-Dahlqvist S, de Jong BA, Berglin E, et al. Antibodies against cyclic citrullinated peptide and IgA rheumatoid factor predict the development of rheumatoid arthritis. *Arthritis Rheum* 2003; **48:** 2741–49.

122 Nielen MM, van Schaardenburg D, Reesink HW, et al. Specific autoantibodies precede the symptoms of rheumatoid arthritis: a study of serial measurements in blood donors. *Arthritis Rheum* 2004; **50:** 380–86.

123 Gerlag DM, Raza K, van Baarsen LG, et al. EULAR recommendations for terminology and research in individuals at risk of rheumatoid arthritis: report from the Study Group for Risk Factors for Rheumatoid Arthritis. *Ann Rheum Dis* 2012; **71:** 638–41.

124 Mankia K, Di Matteo A, Emery P. Prevention and cure: the major unmet needs in the management of rheumatoid arthritis. *J Autoimmun* 2020; **110:** 102399.

125 Mankia K, Siddle H, Di Matteo A, et al. A core set of risk factors in individuals at risk of rheumatoid arthritis: a systematic literature review informing the EULAR points to consider for conducting clinical trials and observational studies in individuals at risk of rheumatoid arthritis. *RMD Open* 2021; **7:** e001768.

126 Mankia K, Siddle HJ, Kerschbaumer A, et al. EULAR points to consider for conducting clinical trials and observational studies in individuals at risk of rheumatoid arthritis. *Ann Rheum Dis* 2021; **80:** 1286–98.

127 Duquenne L, Hensor EM, Wilson M, et al. Predicting inflammatory arthritis in at-risk individuals: development of scores for risk stratification. *Ann Intern Med* 2023; **176:** 1027–36.

128 Van der Helm-van Mil AHM. Preventive interventions in individuals at risk for rheumatoid arthritis: state of the art and perspectives. *Joint Bone Spine* 2023; **90:** 105543.

129 Emery P, Durez P, Dougados M, et al. Impact of T-cell costimulation modulation in patients with undifferentiated inflammatory arthritis or very early rheumatoid arthritis: a clinical and imaging study of abatacept (the ADJUST trial). *Ann Rheum Dis* 2010; **69:** 510–16.

130 Bos WH, Dijkmans BA, Boers M, van de Stadt RJ, van Schaardenburg D. Effect of dexamethasone on autoantibody levels and arthritis development in patients with arthralgia: a randomised trial. *Ann Rheum Dis* 2010; **69:** 571–74.

131 Gerlag DM, Safy M, Maijer KI, et al. Effects of B-cell directed therapy on the preclinical stage of rheumatoid arthritis: the PRAIRI study. *Ann Rheum Dis* 2019; **78:** 179–85.

132 Krijbolder DI, Verstappen M, van Dijk BT, et al. Intervention with methotrexate in patients with arthralgia at risk of rheumatoid arthritis to reduce the development of persistent arthritis and its disease burden (TREAT EARLIER): a randomised, double-blind, placebo-controlled, proof-of-concept trial. *Lancet* 2022; **400:** 283–94.

133 Rech J, Ostergaard M, Tascilar K, et al. Abatacept reverses subclinical arthritis in patients with high-risk to develop rheumatoid arthritis—results from the randomized, placebo-controlled ARIAA study in RA-at risk patients. American College of Rheumatology Convergence; Nov 5–9, 2021 (abstr 0455).

134 Cope A, Jasenecova M, Vasconcelos J, et al. OP0130 abatacept in individuals at risk of developing rheumatoid arthritis: results from the arthritis prevention in the pre-clinical phase of RA with abatacept (APIPPRA) trial. *Ann Rheum Dis* 2023; **82** (suppl 1)**:** 86.

135 Deane KD, Striebich C, Feser M, et al. Hydroxychloroquine does not prevent the future development of rheumatoid arthritis in a population with baseline high levels of antibodies to citrullinated protein antigens and absence of inflammatory arthritis: interim analysis of the StopRA trial. American College of Rheumatology Convergence; Nov 10–14, 2022 (abstr 1604).

136 Di Matteo A, Duquenne L, Cipolletta E, et al. Ultrasound subclinical synovitis in anti-CCP-positive at-risk individuals with musculoskeletal symptoms: an important and predictable stage in the rheumatoid arthritis continuum. *Rheumatology (Oxford)* 2022; **61:** 3192–200.

137 Di Matteo A, Mankia K, Duquenne L, et al. Third-generation anti-cyclic citrullinated peptide antibodies improve prediction of clinical arthritis in individuals at risk of rheumatoid arthritis. *Arthritis Rheumatol* 2020; **72:** 1820–28.

138 Rogier C, Wouters F, van Boheemen L, van Schaardenburg D, de Jong PHP, van der Helm-van Mil AHM. Subclinical synovitis in arthralgia: how often does it result in clinical arthritis? Reflecting on starting points for disease-modifying anti-rheumatic drug treatment. *Rheumatology (Oxford)* 2021; **60:** 3872–78.

139 Garcia-Montoya L, Duquenne L, Nam J, et al. POS0947 subclinical synovitis in anti-CCP positive individuals at-risk of RA: how likely is it to resolve? *Ann Rheum Dis* 2023; **82:** 786.

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